A Proton–Electron Cosmology

RealQM and Newtonian Gravitation from a Single Electric-Potential Seed

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Abstract

We sketch **RealUniv**, an exploratory cosmology built from the RealQM / RealNucleus program, in which all matter is Coulombic (charge clouds in ordinary Euclidean space, no strong force, no elementary neutron) and all gravitation is Newtonian. The novelty here is a **single primitive seed**: a small, small-scale fluctuation of the electric *potential* φ_E alone — charge being its Laplacian, $\rho_c = -\varepsilon_0 \nabla^2 \varphi_E$. From it the universe develops in two tiers. **At the small scale**, the Laplacian inflates the potential seed into charge of both signs and of *different spatial extent* — small dense positive (protons) and large diffuse negative (electrons); since a contracted cloud carries more energy, that extent *is* the inertial mass (E/c^2), and Coulomb geometry then builds nuclei (RealNucleus) and atoms (RealQM). **At the large scale**, the *energy* of that charged matter is a mass/inertia density ρ, and it is this ρ — seeded entirely by the charges — that sources gravitation through Poisson, $\rho = \nabla^2 \varphi_g$. Read about the mean, ρ carries both signs, so the gravitational source is two-fold; the perturbations separate into great chunks of positive mass *interfoliated* with negative mass — the cosmic web of filaments and clusters interleaved with under-dense voids, the negative sector acting as a repulsive dark energy. There is thus *no primitive negative mass and no separate gravitational seed*: only the electric potential is seeded, mass is the energy of the charged world, and gravity is emergent. The construction is strictly non-relativistic — Coulomb below, Newton above — so the speed of light c enters only the electromagnetic sector: there is a local speed limit for charges and light, but none for the large-scale gravitational motion of neutral bodies (two galaxies may recede at relative speed $> c$, as observed). We state the picture, note that it *reproduces* the primordial helium fraction as a consistency (with the observed n–p gap E_* and the standard freeze-out supplied, not derived), and set out its load-bearing open problems — chiefly *what sets the nuclear scale*, the same open datum as E_*.

Status. An exploratory proposal, not established cosmology. It departs from the Standard Model and ΛCDM and must eventually be reconciled with them; the open problems of Section 11 are load-bearing. Read it as "what follows *if*."

Keywords: cosmology; two-tier cosmology; proton–electron model of the nucleus; emergent gravitation; dark energy; primordial nucleosynthesis; RealQM.

1 Introduction

Cosmology is unavoidably speculative at its foundations. The standard account — hot Big Bang, cosmic inflation, ΛCDM — fits the data only by invoking ingredients that are inferred rather than seen: an initial singularity, an inflationary epoch, dark matter, and dark energy, none of them observed in a laboratory, each introduced to close a gap. In a subject resting on such foundations it is legitimate, and useful, to ask what a *simple, rational, non-standard* alternative would look like — one built not from new fields and new epochs but from the most ordinary ingredients physics already has: **electron–proton Coulomb interaction and Newtonian gravitation**. This note develops one such model.

The RealQM program formulates quantum mechanics as multiphase 3D continuum mechanics — unit-charge densities on non-overlapping domains separated by free boundaries, minimising the Coulomb energy [1]. Its nuclear extension, RealNucleus, revives the pre-1932 proton–electron picture (neutron = proton + electron) and reproduces nuclear binding from pure Coulomb interactions, with no strong-force postulate [2]. This note asks what cosmology results if the *same* ingredients are taken as the starting point of the universe, together with ordinary Newtonian gravitation.

The answer, which we call **RealUniv**, is a **two-tier** cosmology from **one seed**. The seed is a fluctuation of the electric *potential* (charge being its Laplacian). It shapes the small scale directly (charge → nuclei → atoms, all of RealQM); and the *energy* of that charged matter, being mass, sources an *emergent* Newtonian gravitation that shapes the large scale (the cosmic web, with a repulsive negative-mass sector as dark energy). We keep strictly to what Coulomb and Newton can say, mark the extreme regimes (strong gravity, the highest energies) where more would be needed, and keep the open problems explicit.

2 The machinery: two-valued Poisson, opposite force-signs

Both forces share *one* structure and differ in a single sign. Each is the gradient of a potential obeying Poisson's equation with a two-valued (±) source,

$$\text{electric:} \quad \rho_c = -\varepsilon_0 \nabla^2 \varphi_E, \qquad\qquad \text{gravitational:} \quad \rho_m = \frac{1}{4\pi G} \nabla^2 \varphi_g, \tag{1}$$

the force being $-(\text{charge})\,\nabla\varphi$ in each case. In this cosmology the electric potential is the primitive seed (Section 3) and the gravitational source is the emergent energy of the charged world (Section 6) — but the *machinery* is common. The one structural difference is the relative sign, and with it the force law:

	like sources	unlike sources
electric (charge)	**repel**	attract
gravitational (mass)	**attract**	repel

Concretely there are two independent binary signs — the source $\pm\nabla^2\varphi$ (two-valued) and the force $\mathbf{F} = \pm\nabla\varphi$ (repulsive or attractive for like sources) — and their *four* combinations are exactly the four "charges" the two forces couple to: the repulsive-for-like sign is electromagnetism, its ± source the ±charge of the small scale; the attractive-for-like sign is gravitation, its ± source the ± mass/energy of the large scale. One operator and two signs thus yield {±charge, ±mass}.

This single sign difference generates the two tiers. Because like charges *repel* and unlike attract, matter neutralises into ± pairs, charge cancels, and electromagnetism *screens* — it stays local and small-scale. Because like masses *attract*, mass *clumps* and accumulates, growing without bound into large-scale structure; and because unlike masses repel, positive- and negative-mass regions *separate* into the interfoliated web and voids of Section 6. The scale separation is thus a *consequence* of the force-sign, not a separate assumption: small-scale electromagnetism and large-scale gravitation are one two-valued Poisson structure, read with two signs. (This is the static Poisson level; the electromagnetic sector carries c and radiation, and ±-mass gravitational dynamics has its own subtleties — both belonging to the extreme regimes outside the Coulomb–Newton scope kept here.)

A second asymmetry — in the *direction* the Poisson relation is used — fixes the *scale* of each force. For electromagnetism the *potential* is primary: the seed is a small, *small-scale* (short-wavelength) oscillation of φ_E — not a smooth ripple — and charge is obtained by *differentiating*,

$\rho_c = -\varepsilon_0 \nabla^2 \varphi_E$. The Laplacian is a *local* operator that *magnifies* short wavelengths ($\rho_k \propto k^2 \varphi_k$), so precisely because the seed is small-scale (large k) it is inflated, despite its tiny amplitude, into sharp, small-scale $\pm$ charges. For gravitation the *source* is primary — the mass/energy of the charged world — and the potential is obtained by *integrating*, $\varphi_g = 4\pi G \nabla^{-2} \rho_m$. The inverse Laplacian is a *global* operator (convolution with the $1/r$ Green's function) that *smooths*, suppressing short wavelengths ($\varphi_k \propto \rho_k / k^2$) and building a broad, large-scale potential. So the same Poisson relation, run forward makes small-scale electromagnetism and run backward makes large-scale gravitation: **the Laplacian acts locally and magnifies; the inverse Laplacian acts globally and smooths** — and that is why charge is small and gravity large.

3 The arena and the single seed

The arena is a plain **3D Euclidean coordinate space** — absolute space, simply given, with no physical medium. The only operator needed is the Laplacian ∇^2, and the only primitive is a small-scale fluctuation of the **electric potential** φ_E. Charge is not postulated separately; it is the curvature of that potential,

$$\rho_{\text{charge}} = -\varepsilon_0 \nabla^2 \varphi_E, \tag{2}$$

with two immediate consequences. **Both signs:** ρ_{charge} is positive where φ_E is concave, negative where convex. **Net zero:** $\int \nabla^2 \varphi_E \, dV$ is a surface term that vanishes, so the universe carries **zero net charge** automatically — equal numbers of protons and electrons, with no baryogenesis (and, once the neutron is read as a proton–electron pair, in the far stronger *local* form of Section 4). And because the Laplacian multiplies each Fourier mode by k^2 ($\rho_k \propto k^2 \varphi_k$), a small, *small-scale* (short-wavelength) oscillation of φ_E — not a smooth ripple — becomes a large density contrast: the seed is tiny in amplitude but short in wavelength, and it is the short wavelength (large k) that the Laplacian amplifies into strong charge.

Crucially, **only the electric potential is seeded** (charge being its Laplacian). Mass and gravitation are not primitive; they arise, in Section 6, as the *energy* of the charged structures the seed produces. This is the essential departure from a cosmology of two independent potentials: here there is one seed, one potential, and gravity is a consequence.

4 The small scale: RealQM and RealNucleus

The seed, inflated by the Laplacian, produces charge of both signs and, decisively, of *different spatial extent*: small dense positive charge and large diffuse negative charge. These are the **proton** and the **electron**, and the whole charged world is built from them by Coulomb geometry.

Size is inertial mass. In RealQM a particle is a charge cloud whose size is fixed by the balance of a localisation (kinetic) cost against Coulomb attraction. A *contracted* cloud costs more localisation and Coulomb energy and so carries *more energy*; and inertia is energy content, mass $= E/c^2$ [3]. Hence **small = heavy, large = light**: the proton (small, dense) is heavy, the electron (large, diffuse) is light. "Proton small, electron big" and "proton heavy, electron light" are one fact, not two — a size difference read through E/c^2. (The numerical ratio $m_p/m_e = 1836$ is not derived; it is the open input of Section 11, as it is in every theory.)

No overlap, no self-interaction: self-repulsion is kinetic energy. RealQM charge clouds do not overlap — each electron occupies its own domain with a free boundary — and carry *no self-interaction*: there is no Coulomb self-repulsion term. The role such self-repulsion would play, resisting the collapse of a charge to a point, is carried instead by the *localisation energy* $\frac{1}{2}\int |\nabla\psi|^2$: compressing a cloud sharpens its gradients and costs kinetic energy, so it resists compression and settles at a finite size (localisation cost against Coulomb attraction). A free electron, unconfined, therefore spreads; a bound one takes the size where localisation balances attraction. In RealQM, **self-repulsion appears as kinetic energy**.

Charge is quantized: ± 1 the only possibility. This same structure forces charge to come in identical units — with no monopole and no compact gauge group imported. To exclude an electron's Coulomb energy with *itself*, the "self" must be well defined: there must be an *indivisible* entity whose interior is left out of the Coulomb sum. A divisible continuum of charge would make "self" ambiguous — is half a cloud a self? do the two halves repel? — and the no-self-interaction prescription ill-posed. So removing self-repulsion *requires* indivisible units. An indivisible unit is identified by its *spatial occupation* — it is precisely what fills one domain — and for all electrons, and equally all protons, to be the *same* entity (identity), each such domain must carry the same charge. Discreteness and universality then follow at once: every electron carries one unit -1 and every proton $+1$, the two signs of the seed (Section 3), and no intermediate charge can exist. The single premise is that the no-self-interaction law is one *universal* law (the same "self" for every electron), not an ad-hoc per-blob subtraction; granted that, quantization is a *consequence*, not a postulate. This is where the Standard Model must *import* the result — anomaly cancellation fixes only the $\pm$ *ratio* of charges, and true quantization needs a Dirac monopole or a compact unifying group, both unobserved. What this argument does *not* fix is the *magnitude* of the unit (why the quantum has the size it does — a scale question, sibling to the nuclear scale of Section 11), nor the fractional charges of confined quarks, which lie outside RealQM's no-substructure scope.

Nuclei and atoms, one mechanism at two scales. With a neutron taken as $p+e$, a neutral atom (A, Z) holds A protons and A electrons: $A-Z$ *compact* inside the nucleus and Z *expanded* in orbitals. The electron occupies two size-phases, its extent adapting to the Coulomb well: compact in a deep well (inside a nucleus, ∼fm, ∼MeV) and expanded in a shallow one (atomic, ∼Å, ∼eV). Chemistry and nuclear physics are then one Coulomb free-boundary mechanism at two scales: a molecule is nuclei glued by expanded (valence) electrons; a nucleus is protons glued by compact electrons. RealNucleus reproduces nuclear binding this way — the He-4/deuteron binding ratio comes out 13.1 against an experimental 12.7, from Coulomb geometry alone [2], and light-nucleus fusion (hydrogen burning) appears as a rearrangement of protons and electrons, not the action of a separate force.

Neutrality is local, not merely cosmic. Because a neutron *is* a bound proton–electron pair, every nucleus (A, Z) carries exactly A protons and A electrons, and $N_e = N_p$ is enforced *atom by atom, by construction* — not as a distant global sum. This is a far stronger statement than cosmic neutrality. Global neutrality asks only that the total charge of the universe vanish; it would tolerate vast regions of net positive and net negative charge cancelling only in the grand total. Local neutrality forbids exactly that: it holds at every atom, and local neutrality *implies* the global kind while the converse fails — so the local form is logically much the stronger. It is also the form actually put to the test: laboratory bulk-matter experiments bound the residual per-atom charge to $|q_p + q_e| \lesssim 10^{-21}\, e$, a direct and stringent constraint, whereas cosmic net-charge limits are indirect inferences. In the Standard Model this exact local balance needs two separate ingredients — charge quantization (from anomaly cancellation) and an *assumed* neutral initial condition — with the neutron, a distinct neutral particle, contributing nothing to the charge ledger and its abundance left to weak freeze-out. Here the same balance is automatic twice over: globally from the net-zero seed ($\int \nabla^2 \varphi_E = 0$, Section 3), and locally because the neutron cannot unbalance a count of which it *is* the matched pair. What remains owed is charge *quantization* — why each cloud carries exactly one e — which rides on the discreteness of the RealQM domains, a separate claim.

The nuclear/atomic scale. What sets the two sizes can be read from charge and the proton's finite size R_p alone, with *no mass and no c*. An electron attracted to one proton in vacuum must taper to zero at infinity; the localisation cost then forces a large cloud of size the atomic $a_0 = \hbar^2/me^2 \approx 0.5\,\text{Å}$. An electron caged among protons at a multiphase free boundary need *not* taper to zero — its density hands off to the proton density — so a flat caged density has zero localisation cost, and it sits at the cage (proton) size $R_p \approx$ a few fm, bound by Coulomb (∼MeV at fm). The size ratio is therefore

$$\frac{a_0}{R_p} \;\approx\; \frac{1}{\alpha^2} \;\approx\; 2\times 10^4, \tag{3}$$

so the fine-structure constant appears here as a pure *geometric* ratio of the two scales, $\alpha^2 = R_p/a_0$, the ratio of nuclear to atomic size — not a coupling constant. Empirically the H atom ($\approx a_0 = 0.53\,\text{Å}$) and the deuteron nucleus ($\approx 2.1\,\text{fm}$) give a ratio $\approx 1/25{,}000$, matching $\alpha^2 \approx 1/18{,}800$ to order of magnitude. *Why* the proton is ∼fm — what fixes R_p — remains the central open problem.

5 Inertia is energy; the local speed limit

Throughout, inertia is the energy content: $\mathbf{p} = (E/c^2)\mathbf{v}$, the measured relation of the Penning trap [3], and Newton's second law $\mathbf{F} = \dot{\mathbf{p}}$ governs the response. In the small, the force is electromagnetic and couples to *charge*; in the large, it is gravity and couples to *mass* $= E/c^2$. One mechanics, one inertia, two forces reading two properties of the same matter.

The constant c enters *only* the electromagnetic sector — through the field momentum of a moving charge, Thomson's electromagnetic mass. Consequently there is a **local speed limit**: a charge, or a pulse of light, cannot overtake light *through* the local matter and fields, because feeding it energy only increases its inertia E/c^2 without bound as $v \to c$. But **Newtonian gravitation is c-free** — instantaneous, with no c in $\rho = \nabla^2\varphi_g$ — so it imposes *no* speed limit on the large-scale motion of neutral bodies. Two galaxies may approach, or recede, at relative speed *greater than* c; this is a global relative velocity of gravitating matter, not a local signal, and it violates no local causality (no local frame sees anything pass light through it). The speed limit is thus a property of the RealQM/Maxwell sector, not of gravity — just as in observational cosmology distant galaxies recede faster than light without any local law being broken. (How Newtonian mechanics extends to relatively moving frames — where c and its kinematics would enter — is a separate matter, taken up in *Many-Minds Relativity* [4], and lies outside the non-extreme scope kept here.)

6 The large scale: emergent Newtonian gravitation

The energy of the charged matter — protons, electrons, their binding and radiation — is a mass/inertia density ρ, **positive throughout** ($\rho \geq 0$), seeded entirely by the charges. It is this positive ρ that sources gravitation, and the emergent potential φ_g comes in **two forms**, its Laplacian taking either sign against the one positive source:

$$\pm\frac{1}{4\pi G}\nabla^2\varphi_g \;=\; \rho \;\geq\; 0. \tag{4}$$

Thus φ_g is not a primitive but *emergent*, the large-scale field of the aggregated energy of the charged world, and it comes in two forms **according to the sign of its Laplacian**: the + form is a *well* ($\nabla^2\varphi_g > 0$, effective positive mass, attractive), the − form an *anti-well* ($\nabla^2\varphi_g < 0$, effective negative mass, repulsive). The ± mass is therefore *formal* — the sign carried by the curvature of φ_g, not a negative energy; the source ρ is positive. By the gravitational force-sign, the wells attract and draw together while the anti-wells repel, so the positive-mass regions are **isolated** into great chunks — filaments and clusters — *interfoliated* with the anti-well regions between them: the **cosmic web**, positive-mass nodes and filaments interleaved with voids, the anti-well (formally negative-mass) sector acting as **dark energy**.

Both the positive and the negative mass are emergent — the two forms of the gravitational potential of the one charge-seeded, positive energy ρ — neither a primitive, neither a second seed; they repel and **separate**, the positive isolating into the web and the negative emptying into the voids. Only charge is put in; the whole mass–gravity structure, of both signs, is the emergent large-scale field of that energy. This is the sense in which the largest structures and the smallest share one origin: the small scale is the seed and its charge directly, the large scale the emergent gravity of their energy.

A 2D test: cosmic-web morphology. A mass-conserving 2D simulation of a $\pm$mass field $\rho = \nabla^2\varphi$ from a random potential, evolved as compressible self-gravitating flow with same-sign attraction and opposite-sign repulsion, develops the morphology of observed large-scale structure (Figure 1): intermixed positive- and negative-mass filamentary webs, mass concentrated at nodes, separated by large voids — the voids here *actively driven* by negative-mass repulsion rather than merely evacuated by gravity. A qualitative match (a 2D toy); quantitative tests (void-size distribution, correlation function) remain to be done.

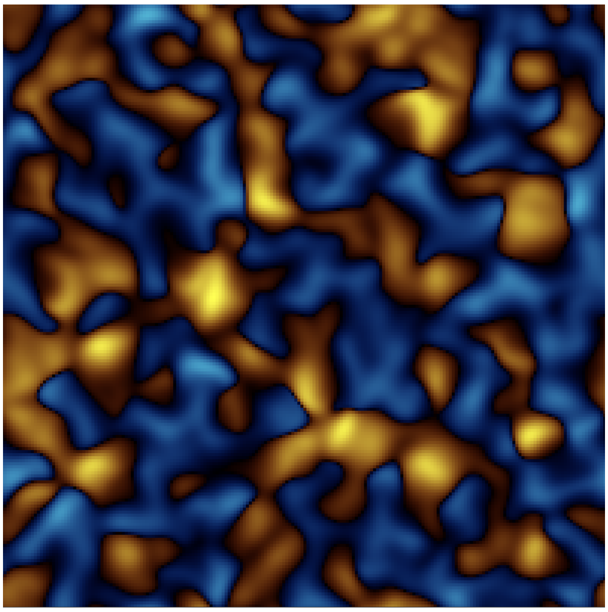

Figure 1: 2D self-gravitating $\pm$mass from $\rho = \nabla^2\varphi$: positive- (warm) and negative- (cool) mass webs with concentrated nodes and large voids — cosmic-web morphology, voids driven by negative-mass repulsion. Mass-conserving; periodic box.

7 Dark energy and inflation from one seed

Two central features of modern cosmology appear here as consequences of the *same* machinery, from the one charge seed — an economy worth stating, and honestly bounding.

Dark energy. The under-dense, formally negative-mass sector of Section 6 is repulsive: the anti-wells push matter away, driving the voids to grow and separate. This is a built-in **dark-energy mechanism**, not put in by hand — it is simply the $-$ side of the density perturbation, the anti-well form of the emergent φ_g. What is *not* yet derived is its magnitude, or that its effective equation of state is $w \approx -1$; the mechanism is qualitative.

That a *negative-mass* sector might unify the dark components has been proposed before, most prominently by Farnes [7], whose single negative-mass fluid — sustained by continuous matter creation — aims to account for both dark matter and dark energy within a modified ΛCDM framework (a proposal that has itself drawn debate). RealUniv shares the core intuition that negative mass drives cosmic acceleration, but differs at the root. There the negative mass is a *primitive* fluid, with genuinely negative inertial mass and runaway motion, requiring a matter-creation term and retaining the expanding-space ΛCDM scaffold; here it is the *emergent, formal* anti-well sign of a gravitational potential sourced entirely by ordinary *positive* energy — no negative inertial mass, no creation term, no Big Bang, and no separate dark-matter component at all (cosmic structure is the ordinary Coulombic mass of Section 6). The two proposals converge on negative mass as the key to the dark sector and diverge on whether it is fundamental or emergent.

They differ, too, in scope. Farnes advances his negative masses avowedly as a *toy model* — a phenomenological component grafted onto ΛCDM to demonstrate that negative mass *can* mimic the dark sector, illustrated by toy N-body simulations. RealUniv instead offers the negative sector as one facet of an *integrated* model built from a single seed: the same charge fluctuation that yields nuclei and atoms (RealQM/RealNucleus) yields the energy that sources gravity, whose anti-well sign *is* the negative sector — derived, not posited, and not bolted onto a standard scaffold. The ambition is thus a full first-principles cosmology rather than a dark-sector patch. Honesty requires the counterweight: on the dark sector *specifically*, RealUniv is so far only qualitative (the 2D toy of Section 6) and *less* quantitatively developed there than Farnes's simulations. RealUniv's claim to be the fuller model rests on integrative scope — one seed for matter and the dark sector alike — not on validated dark-sector dynamics, which remain to be built.

Inflation. The Laplacian amplifies short wavelengths, $\rho_k \propto k^2\varphi_k$, so a tiny small-scale oscillation of the potential is magnified into a large density contrast — an **"inflation" of the density field**. This is amplification of the *field*, not the exponential expansion of *space* of the standard inflationary paradigm (which addresses the horizon and flatness problems); the two must not be conflated.

Because $\rho = \nabla^2\varphi$ is k^2-weighted, the amplification is short-wavelength dominated: the seed comes out as *small-scale* charge structure. In this model that is not a spectral defect to be observed in the sky but the very mechanism by which the initial fluctuation becomes a **proton–electron universe** — nuclei, atoms, bound matter at small scales (Section 4). One might object that a k^2-weighted ("blue") density spectrum conflicts with the near scale-invariant, slightly red primordial spectrum ($n_s \approx 0.96$) inferred from the CMB. But that inference is internal to the hot Big Bang: it reads the CMB as the red-shifted *afterglow* of a hot dense phase and reconstructs a primordial density spectrum through recombination and a transfer function. This model has no Big Bang and no afterglow phase; it makes no claim about the CMB and requires none. The n_s comparison therefore presupposes a framework the model does not adopt, and the short-wavelength weighting is read here as a feature — the origin of small-scale matter — rather than a conflict. The interpretation of the CMB itself we regard as a separate, open observational question outside the scope of a small-scale-structure cosmology.

So from a single seed the model contains natural *candidates* for both dark energy (negative mass) and a fluctuation amplification ("inflation of the density field") — an economy that a separate-mechanism cosmology does not have — while neither is yet quantitatively validated.

8 The cosmic sequence and the helium fraction

The two sectors exchange energy in *both* directions, and the cosmic sequence is that exchange in time. *Upward*, the energy of the charged world is mass and sources gravity (Section 6). *Downward*, gravitational attraction feeds energy back into electromagnetism: the collapse of a positive-mass region converts gravitational potential energy into heat, and that hot dense state drives the electromagnetic processes. Gravity supplies the energy; electromagnetism spends it as structure and light.

Concretely: ordinary gravitational contraction of a positive-mass seed supplies one hot dense epoch, and the released binding energy leaves as light. As the contracting proton–electron matter heats, **deuterons and nuclei** form (∼MeV, RealNucleus), the Coulomb binding energy radiating away (the origin of light) and appearing as the mass defect. On cooling, nuclei capture expanded electrons into **atoms** (∼eV, RealQM), again radiating; at that recombination the matter turns neutral, electromagnetism screens out, and radiation streams free. Only *then* does the weak, slow gravitational aggregation build the cosmic web of Section 6. So the universe's timeline stages the two tiers in order: *charge builds the small while matter is charged; gravity builds the large once it is neutral.*

A bound state survives once the bath can no longer ionise it, so the two electron phases switch on at two temperatures set by their two binding energies, forcing the order plasma → nuclei → atoms (Table 1). The temperature ratio between the epochs ($\sim 10^6$) is just the energy-scale ratio.

epoch	process	temperature
plasma	free ± charges	$> 10^{10}$ K
nuclei	compact e captured → nuclei	$\sim 10^{10}$ K ($kT \sim$MeV)
atoms	expanded e captured → atoms; radiation	$\sim$3000 K ($kT \sim$eV)

Table 1: Two condensation epochs, separated by $\sim 10^6$ in temperature.

The helium fraction: a consistency, not a prediction. The proton–electron picture reproduces the primordial helium; it is worth being exact about what is derived and what is borrowed. The electron has two size-phases — *compact* (inside a nucleus, i.e. inside a neutron, since neutron $= p +$ compact e) and *expanded* (atomic) — interconverted by the two weak processes, compactify ($p + e \to n$) and expand ($n \to p + e$, β-decay). Three quantities enter, and all three are *inputs*:

- E_* — the **energy gap** between the two phases: the energy to compactify an electron, released when it expands. It plays the role the neutron–proton mass difference plays in standard cosmology and is *identified with it*, $E_* \approx 1.29$ MeV. It is an *input*, not a RealNucleus output: its absolute size is the open nuclear-scale problem (Section 11) — RealNucleus fixes dimensionless *ratios*, not the MeV scale.

- T_* — the **transition temperature**, $T_* \sim E_*/k \sim 10^{10}$ K: above it the two phases interconvert freely, below it compactification becomes rare.

- T_f — the **freeze-out temperature** (≈ 0.8 MeV): where the weak (size-changing) processes fall slower than the cosmic expansion and the neutron/proton ratio *locks*. As in standard big-bang nucleosynthesis it is set by the weak-rate-vs-expansion balance, not predicted — a second input.

In equilibrium the neutron/proton ratio is the Boltzmann factor $e^{-E_*/kT}$; at freeze-out it locks at $(n/p)_f = e^{-E_*/kT_f}$; a fraction β-decay back (survival factor d), and essentially all the rest end in He-4:

$$\left(\frac{n}{p}\right)_{\rm nuc} = \frac{(n/p)_f \, d}{1 + (n/p)_f(1-d)}, \qquad Y = \frac{2\,(n/p)_{\rm nuc}}{1 + (n/p)_{\rm nuc}}. \tag{5}$$

With $E_* \approx 1.29$ MeV, $T_f \approx 0.8$ MeV and $d \approx 0.74$: $(n/p)_f = 0.199 \to (n/p)_{\rm nuc} = 0.140 \to Y = 0.245$, the observed primordial helium. This is a **consistency**, not a new prediction: with the two inputs borrowed — E_* the observed n–p gap, T_f the standard freeze-out — the size-transition picture *reproduces* the standard result. Nothing here is derived; and, notably, RealNucleus's one genuine computed output, the He-4/deuteron binding ratio (13.1 vs 12.7), does *not* enter Y. The content is that the proton–electron reinterpretation is *consistent* with the standard helium calculation, with E_* the same open datum as the nuclear scale.

9 Complexity from the size asymmetry

The world is built not on a charge imbalance (charge is exactly balanced, $\pm e$) but on a *size* imbalance: the proton has a fixed small size (a rigid anchor) while the electron's size is variable, adapting to its well — ∼Å in an atom, ∼fm in a nuclear cage. The fixed proton supplies the anchors; the variable electron *spans scales* — spreading to bind atoms and molecules, compacting to bind nuclei — producing the nested multi-scale hierarchy that *is* complexity. With a single fixed size only one scale exists and nothing complex is built. The nuclear/atomic separation is thus a **precondition for complexity**, so any universe that supports structure, and hence observers, must exhibit it: an anthropic reason *why* the scales are separated, though not a derivation of *how far*. Notably, since a neutron is structurally $p + e$, it is heavier than a proton *by construction*, so proton and hydrogen stability is automatic — the usual fine-tuning bound "α must not be too large or the proton decays" simply does not arise.

Why the proton must be a point: free-boundary stability

The size asymmetry is not merely the precondition for complexity; it is what makes an atom a *well-posed, stable* object at all. The decisive quantity is the boundary condition where the electron meets the proton, and it behaves very differently depending on the proton's size.

A RealQM *free boundary* sits where the energy is stationary in its position, and is stable only if that is a minimum. Whether it is depends on the *sign of the force across it.* For *like* charges (electron meeting electron) a displacement raises both the mutual repulsion and the localisation cost — a restoring force, a stable minimum; this is why RealQM's electron–electron boundaries are well behaved. For *opposite* charges (electron meeting proton) the force is *attractive*, and a direct calculation of the cross-interface energy shows it is *monotone* in the separation, pulling the boundary toward contact with no opposing Coulomb term. What "contact" *means* then decides stability, and this turns on the size ratio R_p/a_e of the proton to the electron cloud it meets. If the proton is much smaller, $R_p \ll a_e$ (equivalently $m_p \gg m_e$), it nests as an effective *point* inside the electron: there is no finite interface, "contact" is simply the electron centring on a point charge — the stable, cusped atom — and the electron's localisation cost supplies the restoring force. But if the proton is *comparable*, $R_p \gtrsim a_e$ ($m_p \lesssim m_e$), "contact" means overlap of two comparable clouds, which non-overlap forbids: the Neumann free boundary is pinned against the constraint with no interior minimum, and the configuration collapses. The instability criterion is thus

$$R_p \gtrsim a_e \quad \Longleftrightarrow \quad m_p \lesssim m_e \tag{6}$$

(comparable sizes, order-unity mass ratio). A Neumann-continuous free boundary is the *physical* condition (nothing artificial is imposed), but in this regime it is *unstable.*

One could restore stability with a Dirichlet-0 boundary — a density node between the clouds — but that is *unphysical*: it imposes a node with no physical cause, since nothing in the Coulomb-plus-localisation energy makes the charge vanish at a material interface. So a finite proton is caught in a bind: the *physical* boundary condition (Neumann) is *unstable*, and the *stable* one (Dirichlet) is *unphysical.* A finite p–e interface has no acceptable condition at all.

The resolution nature adopts is to make the proton a *point*: then there is no finite interface, and the dilemma evaporates. The electron simply cusps at the point (maximal density, the Kato cusp) and decays at infinity — the finite-energy, stable ground state — and the free-boundary instability never arises. Minimal spherically symmetric computations bear this out: modelling the proton as a finite charge of radius R_p, both conditions return the exact hydrogen value as $R_p \to 0$ and diverge only for a finite proton (at $R_p = a_0$: the Neumann/static binding is -10.2 eV, the Dirichlet/excluded one only -4.9 eV), while forcing the electron density to zero at the kernel *point* leaves the energy at -13.6 eV unchanged — a node at a single point is measure-zero, and left free the electron closes it and returns to the cusp. Only excluding the electron from a *finite* region costs energy (a fully hollow shell binds at only ≈ -10.6 eV). Thus the size asymmetry is not a convenience but a *necessity*: an equal-sized proton makes the atom either unstable or ill-posed, and matter is well defined only because all the smallness is placed in the proton.

The neutron and the alpha. This same instability reappears at the nuclear scale as a familiar fact. A neutron here is a single $p + e$ pair — one compact electron bound in one proton, with the proton and the compact electron now *comparable* in size: precisely the ill-posed, no-restoring-force regime above. The model therefore predicts the free neutron to be *unstable*, and it is (β-decay, $\sim$15 min); the free-boundary instability *is* the decay — the compact electron expanding back toward atomic size. Nuclei escape it by *collective confinement*. In the α-particle ($^4\text{He} = 4p + 2e$, two compact electrons gluing four protons; Section 4), each electron is caged by several protons at once, so expelling it must work against the surrounding cage — supplying the restoring force a lone pair lacks. The same variational Coulomb minimisation then binds the α (the He-4/deuteron binding ratio comes out 13.1 against an experimental 12.7 [2]), with no strong force. Free neutron unstable, bound neutron stable: one mechanism, matching nature.

10 Relation to ΛCDM and observations

It is worth stating plainly what this model shares with the standard picture, where it differs, and what it does not yet address.

Shared. Ordinary matter is Coulombic and gravitation is attractive between masses; nuclei and atoms form and later assemble into a filamentary large-scale structure of clusters and voids; a helium mass fraction near $Y \approx 0.25$ is accommodated.

Different. There is no initial singularity and no hot Big Bang, hence no cosmic microwave background as a red-shifted afterglow (Section 7); no inflationary expansion of space (the amplification here is of the density *field*, not of space); no dark-matter particle (structure is sourced by the ordinary energy/mass density of Coulombic matter); and no cosmological constant (a repulsive dark-energy sector arises as the negative-curvature part of the emergent gravitational potential). The neutron is composite ($p + e$), not elementary, and nuclear binding is Coulombic geometry rather than a strong force.

Not yet addressed. The model is deliberately restricted to Coulomb+Newton and does not, as it stands, account for the redshift–distance relation and the expansion history, the CMB acoustic-peak spectrum under any alternative interpretation, precise large-scale-structure statistics (the two-point correlation and its baryon acoustic feature), or light-element abundances beyond helium. These are the observational fronts on which any serious successor must eventually be tested; here we claim only internal consistency and economy of ingredients, not empirical parity with ΛCDM. What the model offers in exchange is fewer free primitives — one seed, no dark-matter species, no separate inflaton, no cosmological constant — at the cost of predictive machinery still to be built (Section 11).

11 Open problems

1. **The proton–electron size difference.** This splits into three questions of unequal status. *(i) The ratio.* The electron is $\sim\alpha^{-2}$ larger than the proton, read here as *geometric*: a caged, zero-localisation electron sits at the proton size R_p and a free one at the atomic a_0, so

$a_0/R_p \approx 1/\alpha^2$ (Section 4; no relativity needed). This part the model addresses. *(ii) The absolute scale.* What fixes R_p ($\sim$fm) itself — the same datum as the mass ratio m_p/m_e and the nuclear energy scale: one open number, not derived here (nor in the Standard Model). *(iii) The sign asymmetry.* Why the $+$ sign is the compact one and the $-$ sign the diffuse one *at all*. Pure Coulomb with the Laplacian is charge-conjugation symmetric — swapping $+ \leftrightarrow -$ leaves the equations unchanged — so the asymmetry *cannot* arise from these symmetric ingredients; it requires a symmetry-breaking element. A natural candidate, tied to the fluctuation-spectrum problem below, is a *non-Gaussian (skewed)* potential seed: the super- and sub-level sets of a skewed field are not mirror images — one sign forms sharp, compact pits, the other broad, diffuse swells — a geometric route to one compact and one extended sign from the *statistics* of the seed rather than a new force. This would supply the *direction* of the asymmetry; its *magnitude* is question (ii).

2. **The fluctuation spectrum and a law of motion.** Here $\rho = \nabla^2\varphi$ is a definition, not yet a dynamics: φ needs its own law of motion, and the seed spectrum is an input. The k^2-weighting makes the amplification short-wavelength dominated, which this model reads as the origin of small-scale matter rather than as a primordial-spectrum prediction (Section 7); comparison with the CMB-inferred $n_s \approx 0.96$ presupposes the Big Bang afterglow interpretation, which the model does not adopt. The status of the CMB in a non-Big-Bang setting is itself an open question, but not one this small-scale-structure model needs to settle.

3. **The mass ratio $m_p/m_e = 1836$.** The same open number as the absolute scale, item 1(ii) (small = heavy via E/c^2): built in qualitatively as a size difference, but its value is not derived (nor is it in the Standard Model).

4. **The neutrino.** β-decay's continuous spectrum — the 1930 objection that sank the proton–electron neutron [5]. The "electron expands" step needs a home for that energy and spin.

5. **Precision and scope.** BBN abundances beyond helium, the fluctuation-to-structure statistics, and the extreme regimes (strong-field gravity, the highest energies) lie beyond the Coulomb+Newton scope kept here.

12 Summary

On a given 3D Euclidean space, if the sole primitive is a small-scale fluctuation of the electric *potential* (charge being its Laplacian) — yielding small heavy protons and large light electrons, hence nuclei and atoms by Coulomb geometry (RealQM/RealNucleus) — then the *energy* of that charged matter, single quantity, sources an emergent Newtonian gravitation whose $\pm$ perturbations separate into the cosmic web (positive mass) and dark energy (negative mass), with the speed of light limiting only the electromagnetic sector; a unifying two-tier picture from one seed, whose survival hinges on a single hard question: what sets the nuclear scale.

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